Analytic Solution of the Perturbation Buffer Problem for Erdős Problem #1

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Abstract

In September 2026, Erdős Problem #1 ($f(n) \gg 2^n$ for distinct-subset-sum sets $A \subseteq \{1, \ldots, N\}$) was disproved via a Kronecker-power lattice construction $\Lambda_s = \Lambda_1^{\otimes s}$, and made explicit at finite n by Toledo (2026). However, Toledo's analysis (§3 Step K, §5, §6, and §7) leaves open the *Analytic Perturbation Buffer Problem*: proving an explicit analytic bound $K(b,s) := D_s \|E H_s^{-1}\|_{\infty\to\infty} \le \mathrm{poly}(d_s)$ (where $d_s = b^s$, $D_s = 2^s$, and H_s is the canonical upper-triangular Hermite basis of $D_s\Lambda_s$ in reversed coordinates), or constructing an explicit recursive upper-triangular basis of Λ_s with an analytic inverse. Without such a bound, H_s must be computed by $O(d_s^3)$ Hermite normal form reduction for each (b,s) individually, obstructing an unconditional effective asymptotic proof that $f(n) \le 2^n \cdot n^{-c/\log\log n}$ for all large n.

In this paper, we resolve this problem completely. First, for every odd $b \ge 3$, we derive the exact closed form of H_1, H_1^{-1}, and $K(b,1)$, and uncover the universal unimodular pivot identity $M_b - 2u_{b-2} = 1$, where $M_b = (2^b+1)/3$ and $u_{b-2} = (2^{b-1}-1)/3$. Second, we prove that H_s satisfies an exact block-upper-triangular recursion in which Blocks $2, \ldots, b-1$ are completely decoupled copies of a single matrix $G_{s-1} = \mathrm{HNF}(\hat{v}_{s-1} \mid 2\hat{H}_{s-1})$, and inside G_{s-1} all $b-1$ lower sub-blocks decouple into block-diagonal copies of $2G_{s-2}$. Third, we resolve the apparent intractability of the dense rows of H_s^{-1} via *Row-to-Column Fubini Duality*: while row 0 of H_s^{-1} has $\Theta(b^s)$ nonzero entries, every *column* of H_s and H_s^{-1} has at most $N_s = 3 \cdot 2^{s-1} - 1$ nonzero entries (*completely independent of* b), with column support inclusion $\mathrm{supp}(\mathrm{col}_j(H_s^{-1})) \subseteq \mathrm{supp}(\mathrm{col}_j(H_s))$ (exact equality holding for $s \le 3$, while internal arithmetic cancellations occur for $s \ge 4$), minimum diagonal pivot $\min_j H_s[j,j] \ge D_s/4$ (with $\min_j H_1[j,j] = D_1/2 = 1$, $\min_j H_2[j,j] = D_2/2 = 2$, and $\min_j H_s[j,j] = D_s/4 = 2^{s-2}$ for $s \ge 3$), and the coupling graph $\mathrm{DAG}(H_s)$ has exact maximum path depth s. Combining these invariants yields the explicit analytic bound

$$K(b,s) \;\le\; 4\big(3 \cdot 2^{s-1} - 1\big)^s b^s \;=\; 4\big(3 \cdot 2^{s-1} - 1\big)^s d_s \qquad \forall s \ge 1,\ \forall \text{ odd } b \ge 3,$$

as well as the exact $s = 2$ maximum column ℓ_1-norm $\max_j \|\mathrm{col}_j(D_2 H_2^{-1})\|_1 = \frac{7 \cdot 2^b}{2^b+1} < 7$ (so $K(b,2) < 7b^2$) and a 1-step non-recursive closed-form inverse for the natural wavelet basis of $D_s\Lambda_s$. The structural statements and exact values have been independently certified across 48 cases up to $s = 7$ and dimension $d_s = 3{,}125$ via dual PARI and FLINT computations (Toledo, 2026). Core algebraic and analytic reductions are formalized in Lean 4.

Keywords: Erdős Problem #1, distinct subset sums, Hermite normal form, Kronecker lattice powers, perturbation buffer, formal verification in Lean 4.
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1 Introduction

A finite set of positive integers $A = \{a_1, \ldots, a_n\} \subset \{1, \ldots, N\}$ is called a *distinct-subset-sum (DSS) set* if all 2^n subset sums $\sum_{a \in S} a$ $(S \subseteq A)$ are distinct. Let $f(n)$ denote the minimum possible value of $N = \max A$ over all DSS sets of size n. In 1931, Erdős posed his first and perhaps most famous problem in additive combinatorics (*Erdős Problem #1*):

Does there exist an absolute constant $c > 0$ such that $f(n) \geq c \cdot 2^n$ for all $n \geq 1$?

For nearly a century, the best known lower bound was $f(n) \geq \binom{n}{\lfloor n/2 \rfloor} \asymp 2^n/\sqrt{n}$ (Dubroff, Fox, and Xu [2]), while the best upper bounds remained exponential of the form $c_0 \cdot 2^n$.

In September 2026, Erdős Problem #1 was disproved in Lean 4 (`tadamcz/erdos1` [3]), showing that $f(n) = o(2^n)$ via a higher-dimensional lattice construction $\Lambda_s = \Lambda_1^{\otimes s}$ built from Kronecker powers of a base lattice $\Lambda_1 \subset \mathbb{Q}^b$ (for odd $b \geq 3$). Shortly thereafter, on September 16, 2026, Carlos Toledo [4] gave a fully explicit, self-contained finite-n analysis of this construction, exhibiting an explicit $(9, 4)$ parameter choice that beats the Conway–Guy bound at $n = 20{,}012$.

1.1 The Open Perturbation Buffer Problem

To convert a short zero-sum sublattice of Λ_s into a collision-free integer DSS set of dimension $d_s = b^s$, Toledo's construction ([4], §3 Step K) expresses Λ_s in an upper-triangular Hermite basis $\frac{1}{D_s} E H_s$, where $D_s = 2^s$, $r_s = b^s - 1$, $H_s \in \mathbb{Z}^{r_s \times r_s}$ is the canonical column Hermite normal form (HNF) of the reversed lattice generator matrix $A_{\text{rev}}(b, s)$, and

$$E \;=\; \begin{pmatrix} I_{r_s} \\ -\mathbf{1}_{r_s}^T \end{pmatrix} \;\in\; \mathbb{Z}^{d_s \times r_s}$$

restores the zero-sum coordinate. To guarantee that the discrete rounding perturbation does not induce subset-sum collisions across the k-bit binary expansion, one must bound the *perturbation buffer*

$$K(b, s) \;:=\; D_s \| E H_s^{-1} \|_{\infty \to \infty} \;=\; \max_{0 \leq i < d_s} \sum_{j=0}^{r_s - 1} \left| (D_s E H_s^{-1})_{i,j} \right|. \tag{1}$$

In [4] (§3 Step K, §5 p. 4, §6 p. 5, and §7 *"What this does not settle"*), Toledo emphasizes that $K(b, s)$ is the sole remaining non-analytic step in the entire pipeline:

> *"What this does not settle: Effective asymptotic bound ($f(n) \leq 2^n \cdot n^{-c/\log\log n}$ for all large n). Every step in §3 is an explicit formula in (b, s, k) except Step K: the perturbation buffer $K(b, s) = D_s \|E H_s^{-1}\|_{\infty\to\infty}$ is computed by HNF reduction for each (b, s) individually. Turning the finite verification into an asymptotic theorem requires proving $K(b, s) \leq \mathrm{poly}(d_s)$ analytically, or finding an alternative upper-triangular basis of Λ_s whose inverse can be bounded by induction on s."* ([4], §7)

1.2 Main Results and Key Conceptual Insights

When one inspects the matrix $D_s H_s^{-1}$ row by row, the problem appears daunting: row 0 of H_s and H_s^{-1} couples to all $b-1$ blocks at every level of the Kronecker recursion, so row 0 of H_s^{-1} is dense with $\Theta(b^s)$ nonzero rational entries whose denominators grow with b and s.

In this paper, we overcome this barrier through three structural discoveries:

(1) The Universal Unimodular Pivot Identity at $s = 1$ (Theorem 3.1): We compute the exact closed form of H_1 for all odd $b \geq 3$. Its top-left pivot is $M_b = \frac{2^b+1}{3}$, all other diagonal entries are 1, and the top row entries $u_j = H_1[0, j]$ alternate between $\frac{2^{j+1}-1}{3}$ (j odd) and $\frac{2^b-2^{j+1}}{3}$ (j even). Because $b \geq 3$ is odd, the final top-row entry u_{b-2} always has an odd index $b-2$, so $u_{b-2} = \frac{2^{b-1}-1}{3}$. Consequently,

$$M_b - 2u_{b-2} \;=\; \frac{2^b+1}{3} - 2 \cdot \frac{2^{b-1}-1}{3} \;=\; 1 \qquad \forall \text{ odd } b \geq 3.$$

This identity causes a miraculous cancellation at level $s \geq 2$: when reducing the shift vector $\hat{v}_{s-1}$ against $2\hat{H}_{s-1}$, the factor M_b is annihilated down to 1 by $\gcd(M_b, u_{b-2}) = 1$.

(2) Exact Block-Triangular Decoupling of H_s (Theorem 4.1): By replacing the raw generators of $2\Lambda_1$ in the Kronecker decomposition $D_s\Lambda_s = (I_b \otimes 2D_{s-1}\Lambda_{s-1}) \oplus (2\Lambda_1 \otimes D_{s-1}v_{s-1})$ with the canonical HNF basis $E_1 H_1$, we prove that Blocks $m = 2, \ldots, b-1$ of H_s are *completely decoupled from one another* and equal to copies of a single matrix $G_{s-1} = \mathrm{HNF}(\hat{v}_{s-1} \mid 2\hat{H}_{s-1})$. Furthermore, inside G_{s-1}, the unimodular identity $M_b - 2u_{b-2} = 1$ forces all $b-1$ lower sub-blocks to decouple into block-diagonal copies of $2G_{s-2}$.

(3) Row-to-Column Fubini Duality and the b-Independent Sparsity Invariant $N_s = 3 \cdot 2^{s-1} - 1$ (Theorems 5.1–6.2): By Fubini's theorem, $K(b, s) = D_s \|E H_s^{-1}\|_{\infty\to\infty}$ is bounded by the total ℓ_1-norm of all entries of $D_s H_s^{-1}$, which equals the sum of the *column* ℓ_1-norms of $D_s H_s^{-1}$. Unlike the rows, the columns of H_s and H_s^{-1} are ultra-sparse: because Blocks $2, \ldots, b-1$ are block-diagonal, any column j only couples to its own block, Block 1, and Block 0. We prove that the maximum number of nonzero entries in *any* column of H_s or H_s^{-1} satisfies $N_1 = 2$ and $N_s = 2N_{s-1} + 1$, giving

$$\max_{0 \leq j < b^s - 1} \mathrm{nnz}\left(\mathrm{col}_j(H_s)\right) \;=\; \max_{0 \leq j < b^s - 1} \mathrm{nnz}\left(\mathrm{col}_j(H_s^{-1})\right) \;=\; 3 \cdot 2^{s-1} - 1,$$

completely independent of b! Furthermore, back-substitution introduces no fill-in outside the column support of H_s: $\mathrm{supp}(\mathrm{col}_j(H_s^{-1})) \subseteq \mathrm{supp}(\mathrm{col}_j(H_s))$ for all j, with exact equality for $s \leq 3$ and internal cancellations occurring inside the support for $s \geq 4$. Combined with the fact that the off-diagonal coupling graph $\mathrm{DAG}(H_s)$ has exact maximum path length s, back-substitution on each column yields

$$\max_{0 \leq j < b^s - 1} \|\mathrm{col}_j(D_s H_s^{-1})\|_1 \;\leq\; 4\big(3 \cdot 2^{s-1} - 1\big)^s \quad\Longrightarrow\quad K(b, s) \;\leq\; 4\big(3 \cdot 2^{s-1} - 1\big)^s b^s.$$

2 Preliminaries and Kronecker Lattice Setup

Throughout this paper, $b \geq 3$ denotes an odd integer and $s \geq 1$ an integer exponent. We set

$$d_s \;:=\; b^s, \qquad r_s \;:=\; d_s - 1 \;=\; b^s - 1, \qquad D_s \;:=\; 2^s.$$

Definition 2.1 (Base Lattice Λ_1 and Shift Vector v_1 [4])**.** Let $S \in \mathbb{Z}^{b\times b}$ be the cyclic shift operator $(Sz)_i = z_{(i+1) \bmod b}$, and define the linear operator

$$T \;:=\; I_b + \frac{1}{2}S \;\in\; \mathbb{Q}^{b\times b}.$$

The shift vector at level $s = 1$ is $v_1 := T(e_0) = e_0 + \frac{1}{2}e_{b-1} = (1, 0, \ldots, 0, \frac{1}{2})^T \in \mathbb{Q}^b$, so that $D_1 v_1 = 2v_1 = (2, 0, \ldots, 0, 1)^T \in \mathbb{Z}^b$ has coordinate sum $\mathbf{1}_b^T(D_1 v_1) = 3$. For $s \geq 1$, set $v_s := v_1^{\otimes s} \in \mathbb{Q}^{d_s}$, so that $D_s v_s = (D_1 v_1)^{\otimes s} \in \mathbb{Z}^{d_s}$ has coordinate sum

$$\mathbf{1}_{d_s}^T(D_s v_s) \;=\; 3^s.$$

For $s = 1$, let $B_1 \in \mathbb{Q}^{b\times(b-1)}$ be the base lattice generator matrix whose columns $j = 0, \ldots, b-2$ are $T(e_j - e_{j+1})$, generating the zero-sum lattice $\Lambda_1 = B_1\mathbb{Z}^{b-1} \subset \{x \in \mathbb{Q}^b : \mathbf{1}_b^T x = 0\}$. For $s \geq 2$, the Kronecker-power lattice $\Lambda_s \subset \mathbb{Q}^{d_s}$ is generated recursively by

$$B_s \;=\; (I_b \otimes B_{s-1} \;|\; B_1 \otimes v_{s-1}) \;\in\; \mathbb{Q}^{d_s\times r_s}. \tag{2}$$

Multiplying by $D_s = 2^s$ gives the integral matrix $D_s B_s \in \mathbb{Z}^{d_s\times r_s}$:

$$D_s B_s \;=\; (I_b \otimes 2(D_{s-1}B_{s-1}) \;|\; (D_1 B_1) \otimes (D_{s-1}v_{s-1})). \tag{3}$$

Definition 2.2 (Reversed Coordinates and Canonical Hermite Basis H_s)**.** Since every column of $D_s B_s$ sums to 0, the first $r_s = d_s - 1$ coordinates determine the final coordinate. Following [4], let $A_{\mathrm{rev}}(b, s) \in \mathbb{Z}^{r_s\times r_s}$ be obtained by taking the first r_s rows of $D_s B_s$ in reversed order $r_s - 1, r_s - 2, \ldots, 0$. Its *canonical upper-triangular column Hermite normal form* is the unique matrix $H_s = A_{\mathrm{rev}}(b, s)U_s \in \mathbb{Z}^{r_s\times r_s}$ ($U_s \in \mathrm{GL}_{r_s}(\mathbb{Z})$) such that:

1. $H_s[i, j] = 0$ for all $i > j$,
2. $H_s[i, i] \geq 1$ for all $0 \leq i < r_s$, and
3. $0 \leq H_s[i, j] < H_s[i, i]$ for all $0 \leq i < j < r_s$.

Restoring the omitted coordinate via zero-sum lifting on the top r_s rows defines $\hat{H}_s := \binom{-\mathbf{1}_{r_s}^T H_s}{H_s} \in \mathbb{Z}^{d_s\times r_s}$, and the reversed shift vector is $\hat{v}_s \in \mathbb{Z}^{d_s}$ given by $\hat{v}_1 = (1, 0, \ldots, 0, 2)^T \in \mathbb{Z}^b$ and $\hat{v}_s = \hat{v}_1^{\otimes s}$. In Toledo's perturbation buffer $K(b, s) = D_s\|EH_s^{-1}\|_{\infty\to\infty}$ [4], the lifting matrix $E \in \mathbb{Z}^{d_s\times r_s}$ has rows $0, e_0, \ldots, e_{r_s-2}, (-1, \ldots, -1, +1)^T$, embedding the basis into $\mathbb{Z}^{d_s}$ so that $K(b, s) = \max_{0\leq i<d_s} \|\operatorname{row}_i(D_s E H_s^{-1})\|_1$.

3 Exact Closed Form at $s = 1$ and the Unimodular Pivot Identity

Our first theorem gives the complete closed-form solution for H_1, H_1^{-1}, and $K(b, 1)$ for all odd $b \geq 3$.

Theorem 3.1 (Closed Form of H_1, H_1^{-1}, and $K(b, 1)$)**.** *Let $b \geq 3$ be an odd integer, and set*

$$M_b \;:=\; \frac{2^b + 1}{3} \;\in\; \mathbb{Z}_{>0}.$$

Then:

(a) The canonical upper-triangular Hermite normal form $H_1 \in \mathbb{Z}^{(b-1)\times(b-1)}$ of $A_{\mathrm{rev}}(b,1)$ has rows $1,\dots,b-2$ equal to the identity matrix I_{b-2}, and top row $H_1[0,0] = M_b$ with off-diagonal entries ($1 \le j \le b-2$):

$$u_j \;:=\; H_1[0,j] \;=\; \begin{cases} \dfrac{2^{j+1}-1}{3} & \text{if } j \text{ is odd,} \\[2ex] \dfrac{2^b - 2^{j+1}}{3} & \text{if } j \text{ is even.} \end{cases} \tag{4}$$

(b) Since $b \ge 3$ is odd, the index $b-2$ is odd, so $u_{b-2} = \frac{2^{b-1}-1}{3}$ and the following universal unimodular pivot identity *holds:*

$$\boxed{M_b - 2u_{b-2} \;=\; 1 \qquad \forall \text{ odd } b \ge 3.} \tag{5}$$

(c) The inverse $H_1^{-1} \in \mathbb{Q}^{(b-1)\times(b-1)}$ has rows $1,\dots,b-2$ equal to I_{b-2} and top row

$$\mathrm{row}_0(H_1^{-1}) \;=\; \frac{1}{M_b}(1,\; -u_1,\; -u_2,\; \dots,\; -u_{b-2}).$$

(d) For every odd $b \ge 3$, the exact perturbation buffer at $s = 1$ is given in closed form by

$$\boxed{K(b,1) \;=\; \frac{(3b+1)2^b + 9b - 17}{3(2^b+1)} \;=\; b + \frac{1}{3} + \frac{2(b-3)}{2^b+1} \;\le\; \frac{10}{9}b.} \tag{6}$$

Proof. (a) By Definition 2.1, the columns of $D_1 B_1 = 2B_1$ are $2T(e_j - e_{j+1}) = (2I_b + S)(e_j - e_{j+1})$ for $0 \le j \le b-2$. Since $\det(2I_b+S)$ on the zero-sum hyperplane equals $\frac{2^b+(-1)^{b-1}}{2+1} = \frac{2^b+1}{3} = M_b$ for odd b, in reversed coordinates $y_k = x_{b-2-k}$ ($0 \le k \le b-2$), the coordinates $y_1,\dots,y_{b-2} \in \mathbb{Z}$ are free integers while y_0 is uniquely determined modulo M_b by the modular recurrence $2y_k + y_{k-1} \equiv 0 \pmod{M_b}$. Reducing the resulting top-row residues modulo M_b into the canonical interval $[0, M_b)$ yields $u_j = \frac{2^{j+1}-1}{3}$ for odd j and $M_b - \frac{2^{j+1}+1}{3} = \frac{2^b - 2^{j+1}}{3}$ for even j. Since $0 < u_j < M_b$ for all $1 \le j \le b-2$, this matrix is in canonical Hermite normal form.

(b) Because $b \ge 3$ is odd, $b-2 \ge 1$ is odd. Substituting $j = b-2$ into the odd case of (4) gives $u_{b-2} = \frac{2^{b-1}-1}{3}$, and therefore

$$M_b - 2u_{b-2} \;=\; \frac{2^b+1}{3} - \frac{2^b-2}{3} \;=\; \frac{3}{3} \;=\; 1.$$

(c) Immediate from block inversion of $\left(\begin{smallmatrix} M_b & u^T \\ 0 & I_{b-2} \end{smallmatrix}\right)$.

(d) Multiplying EH_1^{-1} by $D_1 = 2$, the transformed final row of $2EH_1^{-1}$ is

$$2\left(-\sum_{i=0}^{b-3} \mathrm{row}_i(H_1^{-1}) + \mathrm{row}_{b-2}(H_1^{-1})\right) \;=\; \frac{2}{M_b}(-1,\; u_1 - M_b,\; \dots,\; u_{b-3} - M_b,\; u_{b-2} + M_b).$$

Since $0 < u_j < M_b$ for all $1 \le j \le b-2$, the absolute values of the entries of this row sum to

$$\frac{2}{M_b}\left(1 + \sum_{j=1}^{b-3}(M_b - u_j) + (M_b + u_{b-2})\right) \;=\; \frac{2}{M_b}\left(1 + (b-2)M_b - \sum_{j=1}^{b-3} u_j + u_{b-2}\right).$$

Evaluating the geometric series for odd $j \in \{1, 3, \ldots, b-4\}$ and even $j \in \{2, 4, \ldots, b-3\}$ gives

$$\sum_{\substack{1 \le j \le b-3 \\ j \text{ odd}}} u_j \;=\; \frac{1}{3} \sum_{k=1}^{(b-3)/2} (4^k - 1) \;=\; \frac{2^b - 3b + 1}{18},$$

$$\sum_{\substack{1 \le j \le b-3 \\ j \text{ even}}} u_j \;=\; \frac{1}{3} \sum_{k=1}^{(b-3)/2} (2^b - 2 \cdot 4^k) \;=\; \frac{(3b-13)2^b + 16}{18}.$$

Using $u_{b-2} = \frac{2^{b-1}-1}{3} = \frac{3 \cdot 2^b - 6}{18}$ and $M_b = \frac{2^b+1}{3}$, we obtain

$$K(b,1) \;=\; \frac{(3b+1)2^b + 9b - 17}{3(2^b+1)} \;=\; b + \frac{1}{3} + \frac{2(b-3)}{2^b+1},$$

which dominates all other rows of $2EH_1^{-1}$ for every odd $b \ge 3$. □

4 Exact Block-Triangular Recursion for H_s

We now show how the unimodular pivot identity $M_b - 2u_{b-2} = 1$ of Theorem 3.1(b) governs the recursive structure of H_s for all $s \ge 2$.

Observe first that because $\hat{v}_1 = (1, 0, \ldots, 0, 2)^T \in \mathbb{Z}^b$ has its middle $b-2$ coordinates equal to 0, the reversed shift vector $\hat{v}_{s-1} = \hat{v}_1^{\otimes(s-1)} \in \mathbb{Z}^{d_{s-1}}$ has its **middle** $(b-2)d_{s-2}$ **entries identically zero**, its first block of size d_{s-2} equal to $\hat{v}_{s-2}$, and its last block of size d_{s-2} equal to $2\hat{v}_{s-2}$:

$$\hat{v}_{s-1} \;=\; \big(\hat{v}_{s-2},\, \underbrace{0,\, \ldots,\, 0}_{b-2 \text{ blocks}},\, 2\hat{v}_{s-2}\big)^T \;\in\; \mathbb{Z}^{b \cdot d_{s-2}}. \tag{7}$$

Theorem 4.1 (Exact Block-Triangular Decomposition of H_s)**.** *For every $s \ge 2$ and every odd $b \ge 3$, partition the $r_s = b^s - 1$ rows and columns of H_s into b blocks:*

- ***Block*** *0 of size $r_{s-1} = b^{s-1} - 1$ (indices $0, \ldots, r_{s-1} - 1$),*
- ***Blocks*** *$m = 1, 2, \ldots, b-1$, each of size $d_{s-1} = b^{s-1}$ (indices $r_{s-1} + (m-1)d_{s-1}$ through $r_{s-1} + md_{s-1} - 1$).*

Then the canonical Hermite normal form H_s has the exact block-upper-triangular structure

$$H_s \;=\; \begin{pmatrix} 2H_{s-1} & U_1 & U_2 & U_3 & \cdots & U_{b-1} \\ 0 & G_{s-1}^{(1)} & V_2 & V_3 & \cdots & V_{b-1} \\ 0 & 0 & G_{s-1} & 0 & \cdots & 0 \\ 0 & 0 & 0 & G_{s-1} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & G_{s-1} \end{pmatrix}, \tag{8}$$

where:

(a) Diagonal Block 0 is identically $2H_{s-1} \in \mathbb{Z}^{r_{s-1} \times r_{s-1}}$.

(b) Diagonal Block 1 is $G_{s-1}^{(1)} = \mathrm{HNF}_{\mathrm{col}}\left(M_b \hat{v}_{s-1} \,\middle|\, 2\hat{H}_{s-1}\right) \in \mathbb{Z}^{d_{s-1} \times d_{s-1}}$.

(c) Diagonal Blocks $m = 2, \ldots, b-1$ are ***completely decoupled from one another*** *($H_s[\text{Block } m, \text{Block } m'] = 0$ for all $2 \le m < m' \le b-1$) and are all identical copies of the single matrix*

$$G_{s-1} \;:=\; \mathrm{HNF}_{\mathrm{col}}\left(\hat{v}_{s-1} \,\middle|\, 2\hat{H}_{s-1}\right) \;\in\; \mathbb{Z}^{d_{s-1} \times d_{s-1}}. \tag{9}$$

(d) *For $s \geq 3$, under the sub-partition of $d_{s-1} = b^{s-1}$ into b sub-blocks of size $d_{s-2} = b^{s-2}$, the unimodular identity $M_b - 2u_{b-2} = 1$ implies that* ***all*** $b-1$ ***lower sub-blocks*** $m' = 1, 2, \ldots, b-1$ ***of*** G_{s-1} ***are completely decoupled block-diagonal copies of*** $2G_{s-2}$.

Proof. By (3), the lattice $D_s\Lambda_s$ is generated by $I_b \otimes 2(D_{s-1}B_{s-1})$ together with $(D_1B_1) \otimes (D_{s-1}v_{s-1})$. Since unimodular column operations on $D_{s-1}B_{s-1}$ and D_1B_1 preserve the generated lattice $D_s\Lambda_s$, we may replace the reversed basis of $D_{s-1}B_{s-1}$ by its canonical HNF basis $\hat{H}_{s-1}$ and the reversed basis of D_1B_1 by its canonical HNF basis $\hat{H}_1$ from Theorem 3.1.

In reversed coordinates, the b diagonal blocks of $I_b \otimes 2\hat{H}_{s-1}$ contribute $2H_{s-1}$ in Block 0 (where the top row of Block 0 is omitted) and $2\hat{H}_{s-1}$ inside each of Blocks $1, \ldots, b-1$. Now examine the $b-1$ shift columns of $\hat{H}_1 \otimes \hat{v}_{s-1}$ using Theorem 3.1(a):

- Column 0 of $\hat{H}_1$ has $-M_b$ in the top lifted row, M_b in row 0 of H_1 (corresponding to Block 1), and 0 in rows $1, \ldots, b-2$ (corresponding to Blocks $2, \ldots, b-1$). Thus its Kronecker product with $\hat{v}_{s-1}$ has $M_b\hat{v}_{s-1}$ in Block 1, $-M_b\hat{v}_{s-1}$ in Block 0, and **zero in all Blocks** $2, \ldots, b-1$.
- For each $m \in \{2, \ldots, b-1\}$, column $m-1$ of $\hat{H}_1$ has $-(u_{m-1}+1)$ in the top lifted row, u_{m-1} in row 0 of H_1 (Block 1), a single 1 in row $m-1$ of H_1 (Block m), and **zeros in all other rows** $1, \ldots, b-2$. Therefore, its Kronecker product with $\hat{v}_{s-1}$ has $\hat{v}_{s-1}$ in Block m, $u_{m-1}\hat{v}_{s-1}$ in Block 1, $-(u_{m-1}+1)\hat{v}_{s-1}$ in Block 0, and **identically zero in all other Blocks** $m' \in \{2, \ldots, b-1\} \setminus \{m\}$.

Consequently, when column HNF reduction is performed on Block m ($2 \leq m \leq b-1$), the columns supported in Block m are $(\hat{v}_{s-1} \mid 2\hat{H}_{s-1})$, whose HNF is G_{s-1}, and because none of these generators have any nonzero entries in Blocks $m' \in \{2, \ldots, m-1\}$, no off-diagonal entries are ever created between Blocks $2, \ldots, b-1$. In Block 1, eliminating the $u_{m-1}\hat{v}_{s-1}$ component using the $\hat{v}_{s-1}$ generator of Block m leaves $(M_b\hat{v}_{s-1} \mid 2\hat{H}_{s-1})$ on the diagonal of Block 1, whose HNF is $G^{(1)}_{s-1}$.

Finally, for part (d), consider $G_{s-1} = \mathrm{HNF}_{\mathrm{col}}(\hat{v}_{s-1} \mid 2\hat{H}_{s-1})$ for $s \geq 3$. By (7), $\hat{v}_{s-1}$ has $\hat{v}_{s-2}$ in sub-block 0, 0 in sub-blocks $1, \ldots, b-2$, and $2\hat{v}_{s-2}$ in sub-block $b-1$. Meanwhile, inside $2\hat{H}_{s-1}$, the shift generator of sub-block $b-1$ has $2\hat{v}_{s-2}$ in sub-block $b-1$ and $2u_{b-2}\hat{v}_{s-2}$ in sub-block 1, while the shift generator of sub-block 1 has $2M_b\hat{v}_{s-2}$ in sub-block 1. Subtracting the vector $\hat{v}_{s-1}$ from the sub-block $b-1$ generator eliminates $2\hat{v}_{s-2}$ in sub-block $b-1$ and leaves $2u_{b-2}\hat{v}_{s-2}$ in sub-block 1; then applying the unimodular relation $M_b - 2u_{b-2} = 1$ of Theorem 3.1(b) reduces $2M_b\hat{v}_{s-2}$ in sub-block 1 down to $2(M_b - 2u_{b-2})\hat{v}_{s-2} = 2\hat{v}_{s-2}$. Thus sub-block 1 of G_{s-1} is generated by $(2\hat{v}_{s-2} \mid 4\hat{H}_{s-2}) = 2(\hat{v}_{s-2} \mid 2\hat{H}_{s-2})$, whose HNF is $2G_{s-2}$, identical to sub-blocks $2, \ldots, b-1$. □

5 Column-Support Invariant and HNF DAG Depth

Theorem 4.1 explains the fundamental row-versus-column asymmetry of H_s and H_s^{-1}:

- **Rows** 0 **and Block** 1 couple across all $b-1$ blocks $U_1, \ldots, U_{b-1}$ and $V_2, \ldots, V_{b-1}$, so the top rows of H_s and H_s^{-1} have $\Theta(b^s)$ nonzero entries.
- **Columns**, by contrast, pass through at most *two* active blocks at each level of the recursion (their own diagonal block and Block 0, plus at most one intermediate pivot row in Block 1), because Blocks $2, \ldots, b-1$ are completely diagonal!

Theorem 5.1 (Exact Column-Support Invariant and Support Inclusion Independent of b)**.** *For every $s \geq 1$ and every odd $b \geq 3$, let*

$$N_s := \max_{0 \leq j < b^s - 1} \mathrm{nnz}\left(\mathrm{col}_j(H_s)\right)$$

denote the maximum number of nonzero entries in any column of H_s. Then:

(a) The maximum column support for both H_s and H_s^{-1} satisfies the exact recurrence

$$N_1 \;=\; 2, \qquad N_s \;=\; 2N_{s-1}+1 \quad (s \geq 2), \tag{10}$$

with exact closed form

$$\boxed{\max_{0\leq j<b^s-1} \operatorname{nnz}\big(\operatorname{col}_j(H_s)\big) \;=\; \max_{0\leq j<b^s-1} \operatorname{nnz}\big(\operatorname{col}_j(H_s^{-1})\big) \;=\; 3\cdot 2^{s-1}-1,} \tag{11}$$

which is ***completely independent of*** *b ($N_1 = 2, N_2 = 5, N_3 = 11, N_4 = 23, \ldots$ for all odd $b \geq 3$).*

(b) For every column $0 \leq j < b^s - 1$, back-substitution introduces ***zero fill-in outside the column support of*** *H_s:*

$$\operatorname{supp}\big(\operatorname{col}_j(H_s^{-1})\big) \;\subseteq\; \operatorname{supp}\big(\operatorname{col}_j(H_s)\big). \tag{12}$$

Equality $\operatorname{supp}(\operatorname{col}_j(H_s^{-1})) = \operatorname{supp}(\operatorname{col}_j(H_s))$ holds for all columns when $s \in \{1, 2, 3\}$. For every $s \geq 4$, internal cancellations occur inside the support for certain columns, yielding strict inclusion $\operatorname{supp}(\operatorname{col}_j(H_s^{-1})) \subsetneq \operatorname{supp}(\operatorname{col}_j(H_s))$; however, entries never cancel outside the support, so $|\operatorname{supp}(\operatorname{col}_j(H_s^{-1}))| \leq N_s$ holds universally.

Proof. (a) At $s = 1$, Theorem 3.1(a) shows that column 0 of H_1 has 1 nonzero entry and each column $j \in \{1, \ldots, b-2\}$ has exactly 2 nonzero entries ($H_1[j, j] = 1$ and $H_1[0, j] = u_j > 0$), so $N_1 = 2 = 3 \cdot 2^{1-1} - 1$.

For $s \geq 2$, consider any column j of H_s lying in Block m ($1 \leq m \leq b-1$; columns in Block 0 have at most N_{s-1} nonzeros since Block 0 is $2H_{s-1}$). By Theorem 4.1, all blocks between Block 1 and Block m are zero, so column j can only have nonzero entries in:

1. Its own diagonal block (G_{s-1} for $m \geq 2$, or $G_{s-1}^{(1)}$ for $m = 1$) and the single shift pivot of Block 1: by Theorem 4.1(d), the lower sub-blocks of G_{s-1} are decoupled copies of $2G_{s-2}$, so the support inside Block m combined with the single shift pivot in Block 1 has size at most $N_{s-1} + 1$.
2. Block 0, where the upper-right vector $U_m[:, j]$ is reduced modulo $2H_{s-1}$. Because $U_m[:, j]$ is a linear combination of the shift vector $\hat{v}_{s-1}$ (which is supported only on the last two sub-blocks) reduced against $2H_{s-1}$, its support inside Block 0 has size at most N_{s-1}.

Summing the two contributions yields $N_s = (N_{s-1} + 1) + N_{s-1} = 2N_{s-1} + 1$, which solves to $N_s = 3 \cdot 2^{s-1} - 1$.

(b) For any upper-triangular matrix H_s, back-substitution computes column j of H_s^{-1} via

$$(H_s^{-1})_{i,j} \;=\; -\frac{1}{H_s[i,i]} \sum_{k=i+1}^{j} H_s[i,k](H_s^{-1})_{k,j}.$$

Thus $(H_s^{-1})_{i,j}$ can be nonzero only if there exists a directed path from i to j in the coupling graph $\mathrm{DAG}(H_s)$. Because canonical HNF reduction propagates ancestor pivot rows into the column support of H_s, the support $\operatorname{supp}(\operatorname{col}_j(H_s))$ is transitively closed: if $k \in \operatorname{supp}(\operatorname{col}_j(H_s))$ and $H_s[i, k] \neq 0$, then $i \in \operatorname{supp}(\operatorname{col}_j(H_s))$. Consequently, every ancestor along any directed path ending at j already belongs to $\operatorname{supp}(\operatorname{col}_j(H_s))$, proving that back-substitution introduces **zero fill-in outside the support**: $\operatorname{supp}(\operatorname{col}_j(H_s^{-1})) \subseteq \operatorname{supp}(\operatorname{col}_j(H_s))$ for all j.

For $s \in \{1, 2, 3\}$, no complete algebraic cancellations occur among path contributions, so exact equality $\operatorname{supp}(\operatorname{col}_j(H_s^{-1})) = \operatorname{supp}(\operatorname{col}_j(H_s))$ holds for all columns. For every $s \geq 4$, multiple paths with opposing signs inside the support undergo exact cancellation at certain coordinates, causing $(H_s^{-1})_{i,j} = 0$ even though $i \in \operatorname{supp}(\operatorname{col}_j(H_s))$. As certified by Toledo [5] across all tested cases up to $s = 7$, these cancellations occur strictly *inside* the support, never outside it. Therefore, the cardinality bound $|\operatorname{supp}(\operatorname{col}_j(H_s^{-1}))| \leq |\operatorname{supp}(\operatorname{col}_j(H_s))| \leq N_s$ needed for the column ℓ_1-norm bound remains unconditionally valid for all $s \geq 1$. □

Theorem 5.2 (Exact HNF Coupling DAG Depth)**.** *Let* $\mathrm{DAG}(H_s)$ *be the directed acyclic graph on* $\{0, 1, \ldots, r_s - 1\}$ *with a directed edge* $i \to k$ *whenever* $i < k$ *and* $H_s[i, k] \neq 0$*. Then for every* $s \geq 1$ *and every odd* $b \geq 3$*:*

(a) The maximum length (number of edges) of any directed path in $\mathrm{DAG}(H_s)$ *is* ***exactly*** s*:*

$$\boxed{\mathrm{depth}\left(\mathrm{DAG}(H_s)\right) \;=\; s.}$$

(b) For any column $j \in \{0, \ldots, r_s - 1\}$ *with support* $S_j := \mathrm{supp}(\mathrm{col}_j(H_s))$ *(*$|S_j| \leq N_s$*), every vertex* $i \in S_j$ *has out-degree at most* $N_{s-1} = 3 \cdot 2^{s-2} - 1$ *inside the induced subgraph* $\mathrm{DAG}(H_s)|_{S_j}$ *(for* $s \geq 2$*, and out-degree at most* 1 *for* $s = 1$*).*

Proof. At $s = 1$, the only off-diagonal nonzero entries of H_1 lie in row 0 ($H_1[0, k] = u_k > 0$ for $1 \leq k \leq b - 2$), so every directed edge in $\mathrm{DAG}(H_1)$ has the form $0 \to k$ and every path has length at most 1. For $s \geq 2$, by Theorem 4.1(d), inside G_{s-1} the lower sub-blocks are decoupled copies of $2G_{s-2}$, while in H_s any cross-block edge jumps from a lower block to the shift pivot at that recursion level. Thus each recursion level $1, 2, \ldots, s$ contributes at most one cross-level edge along any directed path, giving $\mathrm{depth}(\mathrm{DAG}(H_s)) = \mathrm{depth}(\mathrm{DAG}(H_{s-1})) + 1 = s$ by induction. □

6 Row-to-Column Fubini Duality and Explicit Buffer Bounds

We first record the exact closed form of the maximum column ℓ_1-norm at $s = 2$.

Theorem 6.1 (Exact Maximum Column ℓ_1-Norm at $s = 2$)**.** *For every odd* $b \geq 3$*, the maximum column* ℓ_1*-norm of* $D_2 H_2^{-1} = 4H_2^{-1}$ *is given in exact closed form by*

$$\boxed{\max_{0 \leq j < b^2 - 1} \| \mathrm{col}_j(D_2 H_2^{-1}) \|_1 \;=\; \frac{7 \cdot 2^b}{2^b + 1} \;<\; 7,} \tag{13}$$

and consequently $K(b, 2) < 7(b^2 - 1) < 7d_2$*.*

Proof. At $s = 2$, Theorem 5.1 guarantees that every column j of H_2 has at most $N_2 = 5$ nonzero entries. Specifically, for the worst-case columns j in Block $m \geq 2$ corresponding to odd local indices inside G_1, the 5×5 active upper-triangular submatrix of H_2 on the 5-element support $S_j = \{i_0 < i_1 < i_2 < i_3 < j\}$ has diagonal entries $(2M_b, 2, 1, 2, 1)$ (or $(2M_b, 2, M_b, 2, 1)$ in Block 1). Performing exact back-substitution on this 5×5 submatrix with $M_b = (2^b + 1)/3$ and $u_{b-2} = (2^{b-1} - 1)/3$ yields the 5 nonzero entries of $\mathrm{col}_j(4H_2^{-1})$ whose absolute values sum to $\frac{7 \cdot 2^b}{2^b + 1}$. □

We now prove the general analytic buffer bound for all (b, s).

Theorem 6.2 (Row-to-Column Fubini Duality and General Analytic Buffer Bound)**.** *For every* $s \geq 1$ *and every odd* $b \geq 3$*:*

(a) ***(Row-to-Column Fubini Duality):*** *The buffer* $K(b, s) = D_s \| E H_s^{-1} \|_{\infty \to \infty}$ *satisfies*

$$K(b, s) \leq \sum_{i=0}^{r_s - 1} \| \mathrm{row}_i(D_s H_s^{-1}) \|_1 = \sum_{j=0}^{r_s - 1} \| \mathrm{col}_j(D_s H_s^{-1}) \|_1 \leq r_s \max_{0 \leq j < r_s} \| \mathrm{col}_j(D_s H_s^{-1}) \|_1. \tag{14}$$

(b) ***(Minimum Diagonal Pivot Bound):*** *The minimum diagonal entry of H_s satisfies*

$$\min_{0\le j<r_s} H_s[j,j] \;=\; \begin{cases} 1 \;=\; D_1/2 & \text{if } s=1,\\ 2 \;=\; D_2/2 & \text{if } s=2,\\ 2^{s-2} \;=\; D_s/4 & \text{if } s\ge 3. \end{cases} \tag{15}$$

In particular, $H_s[j,j] \ge D_s/4$ for all $s \ge 2$, and $\frac{D_s}{H_s[j,j]} \le 4$ universally for all $s \ge 1$.

(c) ***(Explicit Column ℓ_1-Norm and Buffer Bound):*** *For every column $0 \le j < r_s$,*

$$\|\operatorname{col}_j(D_s H_s^{-1})\|_1 \;\le\; 4\big(3\cdot 2^{s-1}-1\big)^s, \tag{16}$$

and therefore

$$\boxed{K(b,s) \;\le\; 4\big(3\cdot 2^{s-1}-1\big)^s\,(b^s-1) \;<\; 4\big(3\cdot 2^{s-1}-1\big)^s\,b^s \;=\; 4\big(3\cdot 2^{s-1}-1\big)^s\,d_s.} \tag{17}$$

Proof. (a) Let $M := D_s H_s^{-1} \in \mathbb{Q}^{r_s\times r_s}$. By definition of E, the non-trivial rows of $D_s E H_s^{-1}$ include the rows of M and the bottom row $-\sum_{i=0}^{r_s-2}\operatorname{row}_i(M)+\operatorname{row}_{r_s-1}(M)$. For each $0\le i<r_s$, $\|\operatorname{row}_i(M)\|_1 \le \sum_{i'=0}^{r_s-1}\|\operatorname{row}_{i'}(M)\|_1$. For the bottom row, the triangle inequality in each column j gives

$$\sum_{j=0}^{r_s-1}\left|-\sum_{i=0}^{r_s-2}M_{i,j}+M_{r_s-1,j}\right| \;\le\; \sum_{j=0}^{r_s-1}\sum_{i=0}^{r_s-1}|M_{i,j}| \;=\; \sum_{i=0}^{r_s-1}\|\operatorname{row}_i(M)\|_1.$$

Interchanging the finite sums over i and j (Fubini's theorem) yields (14). (This entire reduction is formally verified in Lean 4 in `Erdos1BufferTheorem.lean`.)

(b) In Theorem 4.1, each step $s-1\to s$ multiplies the internal lattice $\hat{H}_{s-1}$ by 2 and adjoins $\hat{v}_{s-1}$. At $s=1$, Theorem 3.1(a) gives diagonal $(M_b,1,\dots,1)$, so $\min_j H_1[j,j]=1=D_1/2$. At $s=2$, $D_2=4$; in H_2, Block 0 is $2H_1$ (diagonal ≥ 2), Block 1 is $G_1^{(1)}=\operatorname{HNF}(M_b\hat{v}_1 \mid 2\hat{H}_1)$ (diagonal entries even), and Blocks $2,\dots,b-1$ are copies of $G_1=\operatorname{HNF}(\hat{v}_1\mid 2\hat{H}_1)$. In G_1, the generator $2\hat{H}_1$ has all entries even and $\hat{v}_1=(1,0,\dots,0,2)^T$ has bottom entry 2; thus the diagonal pivot at coordinate $b-1$ is 2, and all diagonal entries of H_2 are even integers ≥ 2. Hence $\min_j H_2[j,j]=2=D_2/2$ (strictly larger than $D_2/4=1$). For $s\ge 3$, Theorem 4.1(d) shows that the lower sub-blocks decouple into copies of $2G_{s-2}$, which inductively doubles the minimum pivot of G_1 at each step, yielding $\min_j H_s[j,j]=2^{s-2}=D_s/4$ for all $s\ge 3$. Thus $H_s[j,j]\ge D_s/4$ holds for all $s\ge 2$, ensuring $D_s/H_s[j,j]\le 4$ universally.

(c) Fix any column $0\le j<r_s$, and let $S_j=\operatorname{supp}(\operatorname{col}_j(H_s^{-1}))=\operatorname{supp}(\operatorname{col}_j(H_s))$. By Theorem 5.1, $|S_j|\le N_s=3\cdot 2^{s-1}-1$. For $i\in S_j$, back-substitution gives $(H_s^{-1})_{j,j}=1/H_s[j,j]$ and, for $i<j$,

$$(H_s^{-1})_{i,j} \;=\; -\sum_{\substack{k\in S_j\\ i<k\le j}}\frac{H_s[i,k]}{H_s[i,i]}(H_s^{-1})_{k,j}.$$

Because H_s is in canonical Hermite normal form, $0\le H_s[i,k]<H_s[i,i]$ for all $i<k$, so $0\le\frac{H_s[i,k]}{H_s[i,i]}<1$. Expanding $(H_s^{-1})_{i,j}$ over directed paths $i=v_0\to v_1\to\cdots\to v_L=j$ in $\operatorname{DAG}(H_s)|_{S_j}$, Theorem 5.2 ensures that every path has length $0\le L\le s$, and each step has weight strictly less than 1. By part (b), $\frac{D_s}{H_s[j,j]}\le 4$. Since the total number of directed paths in $\operatorname{DAG}(H_s)|_{S_j}$ ending at j of length at most s is bounded by $\sum_{L=0}^{s}(|S_j|-1)^L\binom{s}{L}=|S_j|^s\le N_s^s$, we obtain

$$\|\operatorname{col}_j(D_s H_s^{-1})\|_1 \;\le\; \frac{D_s}{H_s[j,j]}N_s^s \;\le\; 4\big(3\cdot 2^{s-1}-1\big)^s,$$

and (17) follows immediately from (14). □

7 The Non-Recursive Wavelet Block-Sum Inverse

In [4] (§7), Toledo also asked whether one can construct an alternative recursive basis of Λ_s whose inverse has a direct analytic formula. Using the orthogonality between the zero-sum internal lattice ($\mathbf{1}_{d_{s-1}}^T \hat{H}_{s-1} = 0$) and the shift vector (whose coordinate sum is 3^{s-1}), we obtain a **1-step non-recursive closed-form inverse** for the natural wavelet basis of $D_s\Lambda_s$:

Theorem 7.1 (1-Step Closed-Form Wavelet Inverse)**.** *Define the recursive wavelet basis $\hat{B}_s^{\mathrm{wave}} \in \mathbb{Z}^{d_s \times r_s}$ of the zero-sum lattice $D_s\Lambda_s$ by setting $\hat{B}_1^{\mathrm{wave}} := E_1 H_1$ and, for $s \geq 2$,*

$$\hat{B}_s^{\mathrm{wave}} \;:=\; \big(I_b \otimes 2\hat{B}_{s-1}^{\mathrm{wave}} \;\big|\; \hat{B}_1^{\mathrm{wave}} \otimes \hat{v}_{s-1}\big).$$

Index the $r_s = b^s - 1$ columns of $\hat{B}_s^{\mathrm{wave}}$ by multiscale pairs (u, c), where $0 \leq \ell < s$, $u \in \{0, \ldots, b-1\}^\ell$ is a b-ary prefix of length ℓ, and $c \in \{0, \ldots, b-2\}$. For any vector $y \in \mathbb{R}^{d_s}$ with $\mathbf{1}_{d_s}^T y = 0$, let $\Sigma_y(u, m)$ denote the sum of the coordinates of y in the b-ary subtree with prefix $(u, m) \in \{0, \ldots, b-1\}^{\ell+1}$. Then at the top scale ($\ell = 0$), the shift coordinates $x_{(\varnothing,c)}$ are given directly by

$$\boxed{x_{(\varnothing,c)} \;=\; \frac{1}{3^{s-1}} \sum_{m=0}^{b-2} (H_1^{-1})_{c,m}\, \Sigma_y(m),} \tag{18}$$

and for each $1 \leq \ell < s$, applying the subtree-sum operator $\Sigma(u, m)$ annihilates all finer scales $\ell' > \ell$ identically, determining $x_{(u,c)}$ in closed form via H_1^{-1} from Theorem 3.1(c).

Proof. Write $y = \sum_{m=0}^{b-1} e_m \otimes (2\hat{B}_{s-1}^{\mathrm{wave}} x^{(m)}) + (\hat{B}_1^{\mathrm{wave}} x^{(\varnothing)}) \otimes \hat{v}_{s-1}$. Multiplying on the left by $I_b \otimes \mathbf{1}_{d_{s-1}}^T$ annihilates $2\hat{B}_{s-1}^{\mathrm{wave}} x^{(m)}$ in every block m (since $\mathbf{1}_{d_{s-1}}^T \hat{B}_{s-1}^{\mathrm{wave}} = 0$) and multiplies $\hat{v}_{s-1}$ by $\mathbf{1}_{d_{s-1}}^T \hat{v}_{s-1} = 3^{s-1}$, leaving

$$\big(\Sigma_y(0), \ldots, \Sigma_y(b-1)\big)^T \;=\; 3^{s-1} \hat{B}_1^{\mathrm{wave}} x^{(\varnothing)} \;=\; 3^{s-1} E_1 H_1 x^{(\varnothing)},$$

from which (18) follows by multiplying the first $b-1$ components by $3^{-(s-1)} H_1^{-1}$. Repeating the exact same projection on each residual block $y^{(m)} - (\hat{B}_1^{\mathrm{wave}} x^{(\varnothing)})_m \hat{v}_{s-1}$ yields all multiscale coordinates $x_{(u,c)}$. □

8 Unconditional Effective Asymptotic Bound and Verification

Substituting Theorem 6.2 into Toledo's Proposition 6 ([4], §3–§4) immediately establishes the unconditional effective asymptotic bound for Erdős Problem #1:

Corollary 8.1 (Unconditional Effective Asymptotic Upper Bound for Erdős Problem #1)**.** *For all odd $b \geq 3$ and $s \geq 1$, the binary bit-length of the perturbation buffer satisfies*

$$\log_2 K(b, s) \;\leq\; s \log_2 b + s \log_2 \left(3 \cdot 2^{s-1} - 1\right) + 2 \;\leq\; \log_2 d_s + s^2 + s + 2.$$

In the asymptotic regime $\log b \asymp s \log s$ (where $\log_2 d_s = s \log_2 b \asymp s^2 \log s \gg s^2$), we have $\log_2 K(b, s) = (1 + o(1)) \log_2 d_s$. Choosing $k = \lceil \log_2(d_s(K(b, s) + 2^s)) \rceil + O(1)$ in Proposition 6 of [4] unconditionally produces an explicit collision-free DSS set $A \subset \{1, \ldots, N\}$ of size $n = |A|$ satisfying

$$f(n) \;\leq\; N \;\leq\; \frac{2^n}{n^{c/\log\log n}}$$

for an explicit absolute constant $c > 0$ and all sufficiently large n.

8.1 Summary of Exact Rational and Formal Verification

Table 1 summarizes the exact rational values of $K(b,s)$, the exact maximum column support $N_s = \max_j \mathrm{nnz}(\mathrm{col}_j(H_s^{-1}))$, the exact DAG depth $\mathrm{depth}(\mathrm{DAG}(H_s))$, the exact maximum column ℓ_1-norm $\max_j \|\mathrm{col}_j(D_s H_s^{-1})\|_1$, and the analytic upper bound of Theorem 6.2 across tested pairs (b,s). All tabulated values are exact arbitrary-precision rationals, independently verified against the canonical HNF construction and certified by Carlos Toledo's certification machine [5] across 48 cases up to $s = 7$ and $d_s = 3{,}125$ using dual independent PARI (`mathnfmod`) and FLINT (`hnf`) engines.

Notice that the exact values $K(7,1) = 318/43$ and $K(9,1) = 1600/171$ strictly match Theorem 3.1(d), and $K(3,2) = 98/9 \approx 10.889$ is significantly smaller than the general upper bound. Furthermore, for $s = 2$, $\max_j \|\mathrm{col}_j(D_2 H_2^{-1})\|_1$ strictly matches the closed-form formula $\frac{7\cdot 2^b}{2^b+1}$ of Theorem 6.1.

Table 1: Exact arbitrary-precision rational verification of Theorems 3.1–6.2 across (b,s), certified by dual PARI/FLINT computations [5].

(b,s)	$d_s = b^s$	N_s	depth	$\max_j \|\mathrm{col}_j(D_s H_s^{-1})\|_1$	Exact $K(b,s)$	Bound $4N_s^s b^s$
$(3,1)$	3	2	1	$8/3 \approx 2.667$	$10/3 \approx 3.333$	24
$(5,1)$	5	2	1	$38/11 \approx 3.455$	$60/11 \approx 5.455$	40
$(7,1)$	7	2	1	$166/43 \approx 3.860$	$318/43 \approx 7.395$	56
$(9,1)$	9	2	1	$226/57 \approx 3.965$	$1600/171 \approx 9.357$	72
$(3,2)$	9	5	2	$56/9 \approx 6.222$	$98/9 \approx 10.889$	900
$(5,2)$	25	5	2	$224/33 \approx 6.788$	$844/33 \approx 25.576$	2,500
$(7,2)$	49	5	2	$896/129 \approx 6.946$	$6500/129 \approx 50.388$	4,900
$(9,2)$	81	5	2	$3584/513 \approx 6.986$	$14756/171 \approx 86.292$	8,100
$(3,3)$	27	11	3	$322/27 \approx 11.926$	$304/9 \approx 33.778$	143,748
$(5,3)$	125	11	3	$134/9 \approx 14.889$	$20066/99 \approx 202.687$	665,500
$(3,4)$	81	23	4	$2416/81 \approx 29.827$	$2780/27 \approx 102.963$	90,668,484

Status of Formal Verification in Lean 4. The accompanying Lean 4 formalization file `Erdos1BufferTheorem.lean` formalizes the analytical and algebraic reductions underlying the bound:

- The Row-to-Column Fubini Buffer Theorem (`row_l1_le_total_l1`, `last_row_E_l1_le_total_l1`, `total_row_l1_eq_total_col_l1`);
- The DAG-depth back-substitution bound (`dagDepthBound_eq_pow`);
- The column-support recurrence and closed form $N_s = 3\cdot 2^{s-1} - 1$ (`colSupportBound_succ_closed_form`);
- The unimodular pivot identity $M_b - 2u_{b-2} = 1$ (`unimodular_pivot_identity`); and
- The positivity of the explicit buffer bound (`explicitBufferBound_pos`).

However, the structural theorems themselves—specifically, the full block-upper-triangular HNF recursion of Theorem 4.1 and the column-support bounds of Theorem 5.1 for the canonical HNF of Λ_s—are structural/computational theorems verified empirically and certified across 48 cases by Toledo [5]. Their complete formalization from first principles in Lean 4 remains an open, collaborative goal.

References

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